

1. Question: What was the primary computing platform characteristic of the First Generation (1950-1970)?

- A) Minicomputers
- B) Personal computers
- C) Mainframes
- D) Portable devices

2. Question: High-Performance Computing (HPC) is primarily focused on raw speed and is measured in what unit?

- A) Cost reduction units
- B) Energy efficiency units
- C) Floating-point operations per second (FLOPS)
- D) Number of simultaneous tasks

3. Question: Which emerging computing paradigm is described as the "evolution of computing as a utility"?

- A) Service-Oriented Architecture (SOA)
- B) Virtualization
- C) Internet of Things (IoT)
- D) Cloud Computing

4. Question: Which degree of parallelism includes techniques such as pipelining, superscalar processing, and multithreading?

- A) Bit-level parallelism (BLP)
- B) Instruction-level parallelism (ILP)
- C) Data-level parallelism (DLP)

D) Task-level parallelism (TLP)

5. Question: In which year was the Internet of Things (IoT) concept first introduced at MIT?

A) 1989

B) 1999

C) 2009

D) 2019

---SUMMARY---

The text outlines the historical evolution of computing from early mainframes to modern scalable and distributed systems, emphasizing the shift from centralized computing to parallel and High-Performance Computing (HPC) and High-Throughput Computing (HTC) paradigms, culminating in cloud computing as a utility. This progression is underpinned by significant advancements in multicore CPUs, massively parallel GPUs, faster memory and storage (including SSDs), and high-bandwidth networking. Key enabling technologies discussed include virtualization, Service-Oriented Architecture (SOA), the Internet of Things (IoT), and Cyber-Physical Systems (CPS), all aimed at achieving greater efficiency, dependability, and adaptability for future, complex, and data-intensive applications.